Reordering & Rearrangment

Secret Revealed

Definition 1. Let $f : \mathbb{N} \to \mathbb{N}$ be a bijective function *or* we can say f is a *permutation* on $\mathbb{N}$. Let $\left(a_n\right)$ be a sequence. Then $\left(a_{f(n)}\right)$ is called a *reordering* of $\left(a_n\right)$.

Definition 2. Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be two series. Then $\sum_{k=1}^{\infty} b_k$ is called a *rearrangement* of $\sum_{k=1}^{\infty} a_k$ if there exists a permutation $f : \mathbb{N} \to \mathbb{N}$ such that $b_{f(k)} = a_k$.

Theorem 1. Any reordering of a convergent sequence converges to the same point.

Let $\left(a_n\right)$ be a convergent sequence. Then $\left(a_{f(n)}\right)$ converges to the same limit (where $f : \mathbb{N} \to \mathbb{N}$ is a permutation), i.e., $\lim\left(a_n\right) = \lim\left(a_{f(n)}\right)$.

Proof. (Q14, WS2)

Theorem 2. Any rearrangement of an absolutely convergent series converges to the same point.

Let $\sum_{k=1}^{\infty} a_k$ be an absolutely convergent series. Then $\sum_{k=1}^{\infty} a_{f(k)}$ converges to the same limit, i.e., $\sum_{k=1}^{\infty} |a_k|$ converges $\Rightarrow \sum_{k=1}^{\infty} a_{f(k)} = \sum_{k=1}^{\infty} a_k$.

Proof. (Q15, WS4)

Example 1: Let $\left(a_n\right)=\left(\frac{(-1)^{n+1}}{n}\right)=\left(1,-\frac{1}{2},\frac{1}{3},-\frac{1}{4},\frac{1}{5},-\frac{1}{6},\ldots\right)$ and

$$\left(b_n\right)=\left(-\frac{1}{2},1,-\frac{1}{4},\frac{1}{3},-\frac{1}{6},\frac{1}{5},\ldots\right) \text{ or } b_n=\begin{cases}\frac{-1}{n+1}, n\equiv 1 \pmod 2\\ \frac{1}{n-1}, n\equiv 0 \pmod 2\end{cases}.$$

Then $\left(b_n\right)$ is a reordering of $\left(a_n\right)$. Note that

$$\lim\left(a_n\right)=0=\lim\left(b_n\right) \text{ (Theorem 1 is verified)}$$

Example 2: Let $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = A_1 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \ldots$

$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{2(n+1)}{n(n+2)} \right) = A_2 = 1 + \frac{1}{3} - \frac{1}{2} - \frac{1}{4} + \frac{1}{5} + \frac{1}{7} - \frac{1}{6} - \frac{1}{8} + \ldots$

$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = B_1 = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \frac{1}{6^2} + \frac{1}{7^2} - \frac{1}{8^2} + \ldots$

$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{2(n^2+2n+2)}{n^2(n+2)^2} \right) = B_2 = 1 + \frac{1}{3^2} - \frac{1}{2^2} - \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{7^2} - \frac{1}{6^2} - \frac{1}{8^2} + \ldots$

Then A_2 is a rearrangement of A_1, and B_2 is a rearrangement of B_1,

Explain whether $A_1 = A_2$ and/or $B_1 = B_2$.

Recall in Chapter 1, in the very *first* lecture…

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \ldots = \ln 2 \approx 0.693\,.$$

However, we arrive at another answer when we rearrange the infinite series as

$$1 + \tfrac{1}{3} - \tfrac{1}{2} + \tfrac{1}{5} + \tfrac{1}{7} - \tfrac{1}{4} + \ldots$$

Summing the first 500 terms gives ≈ 1.038, it is nowhere near $\ln 2$ (or 0.693)!!

Can you explain why?